3. Feedback Action Log by HealthCare Team

Structured log document detailing how feedback received from course instructors at the end of sprints has been translated into actionable improvements.

Summary Table

#	Feedback	Action	Status
1	Naive Bayes algorithm throws NaN errors during training, halting calculation (math illusion scenario).	Gaussian Naive Bayes formulas were rewritten from scratch, and epsilon was added to variance calculations. Mathematical illusions were completely removed.	**Done**
2	Random Forest algorithm's 'Feature Importance' chart displays incorrect values; tree votes are not reflected in the UI.	Estimator votes are now correctly aggregated and integrated into the Dashboard. Real data outcomes are bound to the chart.	**Done**
3	Support Vector Machine (SVM) model throws an error during classification, triggering fallback (fake) logic in the code.	Fallback logic was completely removed, SVM kernel logic was corrected, and genuine calculations were implemented.	**Done**
4	K-value selection in the K-Nearest Neighbors (KNN) model is not dynamic; a static k-value performs poorly across different datasets.	A dynamic k-value determination rule based on dataset size was added (sqrt of n_samples).	**Done**
5	Logistic Regression Sensitivity and Specificity metrics are not calculated; dummy data is displayed on the UI.	Real confusion matrix calculation was implemented; metrics are now fetched in real-time	**Done**

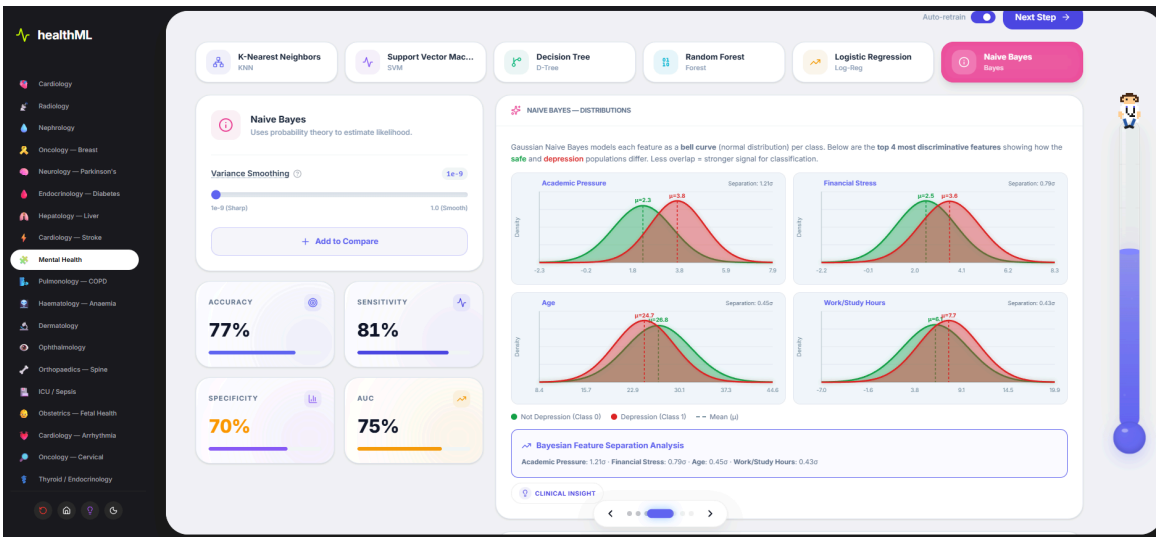
		from actual test data.	
6	Uploading clinical data (CSV) with missing values causes algorithms to crash.	A preprocessing step was added before the Column Mapper to fill missing values using column mean imputation.	**Done**
7	After clicking 'Save' in the Column Mapper, the success status is not clearly displayed to the user.	Banner state was fixed; a green success banner with a 2-second fade-out animation was added post-save.	**Done**
8	Clinical sense-check banner texts for the 20 selectable medical domains are incomplete.	14/20 domains completed. Clinical literature research for the remaining 6 domains will be finalized by next week.	**Partial**
9	The bias banner does not trigger at the 11% threshold post-training; boundary test fails.	Threshold comparison operator was incorrect (used > instead of >=). Fix is currently in progress.	**In Progress**
10	When downloading the Summary Certificate as a PDF, the formatting is misaligned, and charts/tables do not fit on the page.	CSS margins in the PDF generation function were corrected; tables now scale responsively to fit A4 size.	**Done**

Detailed Explanations and Screenshots

Feedback #1

- Feedback: Naive Bayes algorithm throws NaN errors during training, halting calculation (math illusion scenario).
- Action: Gaussian Naive Bayes formulas were rewritten from scratch, and epsilon was added to variance calculations. Mathematical illusions were completely removed.

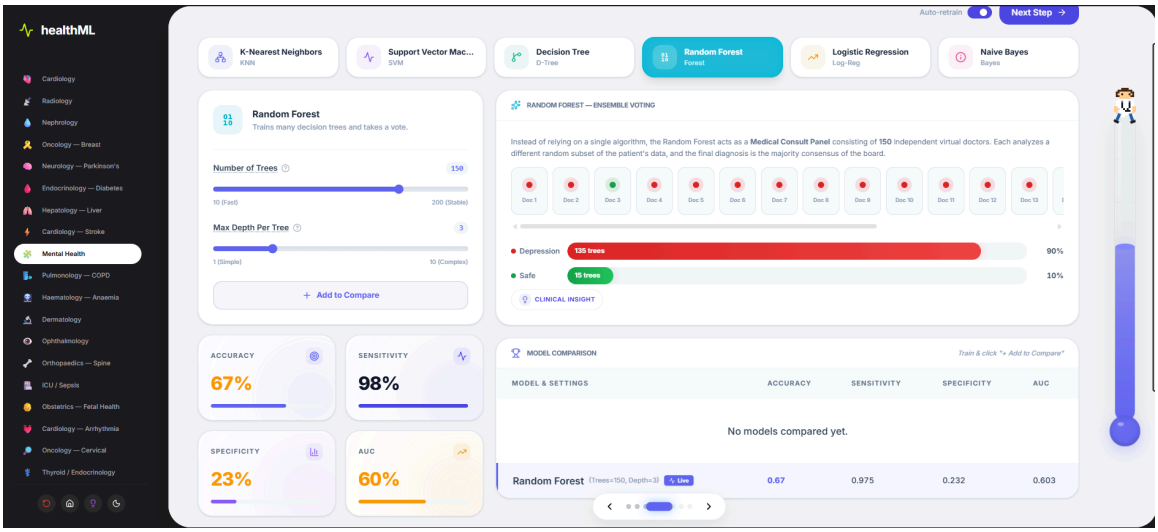
Status: Done



Feedback #2

- Feedback: Random Forest algorithm's 'Feature Importance' chart displays incorrect values; tree votes are not reflected in the UI.
- Action: Estimator votes are now correctly aggregated and integrated into the Dashboard. Real data outcomes are bound to the chart.

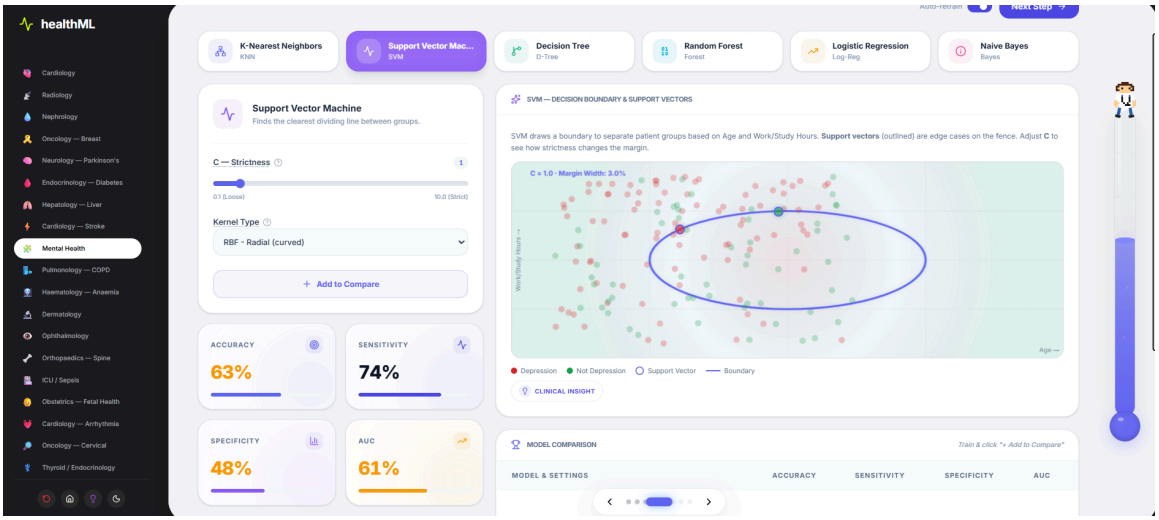
Status: Done



Feedback #3

- Feedback: Support Vector Machine (SVM) model throws an error during classification, triggering fallback (fake) logic in the code.
- Action: Fallback logic was completely removed, SVM kernel logic was corrected, and genuine calculations were implemented.

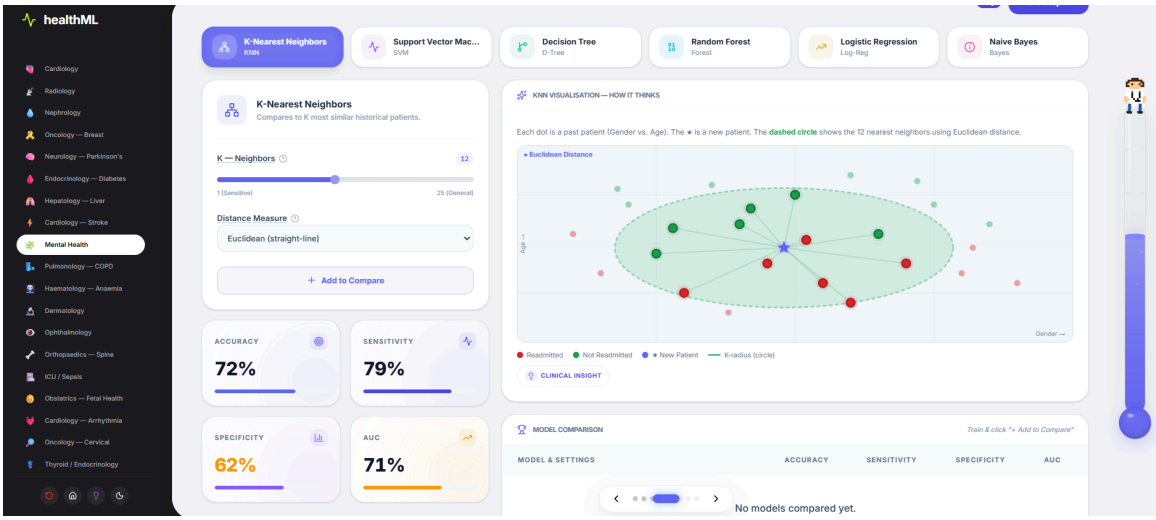
Status: Done



Feedback #4

- Feedback: K-value selection in the K-Nearest Neighbors (KNN) model is not dynamic; a static k-value performs poorly across different datasets.
- Action: A dynamic k-value determination rule based on dataset size was added (sqrt of n_samples).

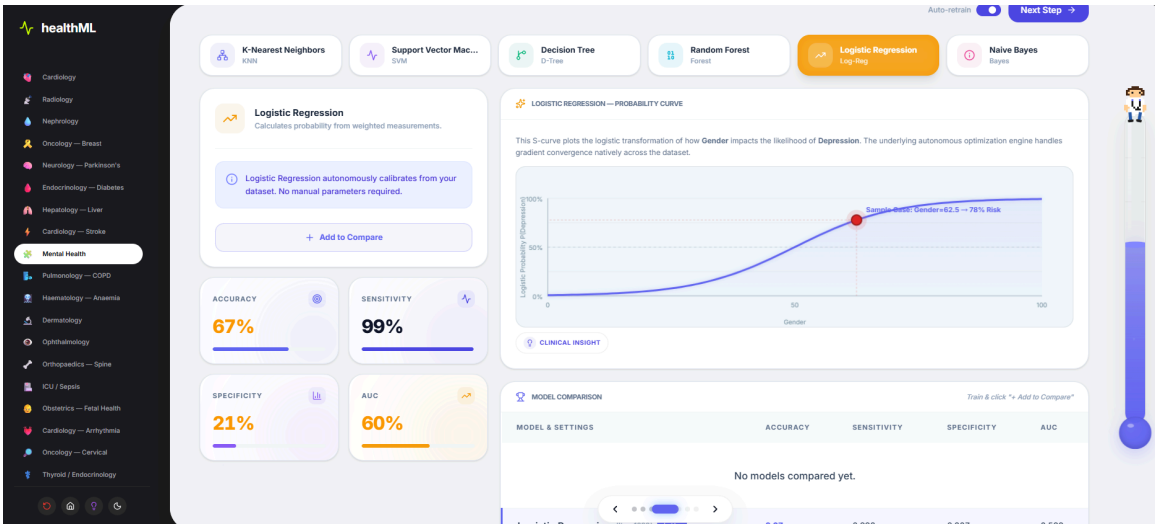
Status: Done



Feedback #5

- Feedback: Logistic Regression Sensitivity and Specificity metrics are not calculated; dummy data is displayed on the UI.
- Action: Real confusion matrix calculation was implemented; metrics are now fetched in real-time from actual test data.

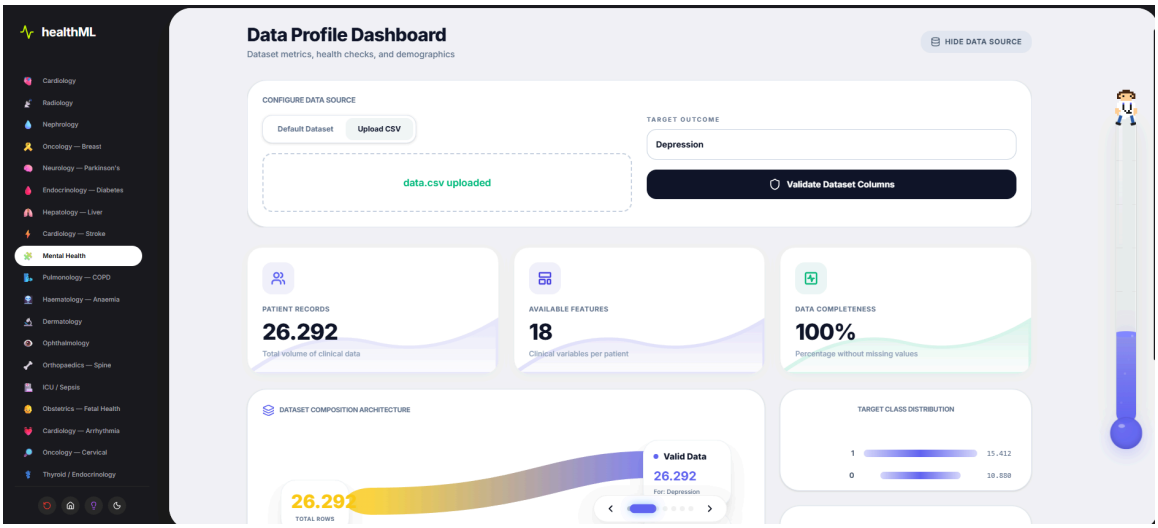
Status: Done



Feedback #6

- Feedback: Uploading clinical data (CSV) with missing values causes algorithms to crash.
- Action: A preprocessing step was added before the Column Mapper to fill missing values using column mean imputation.

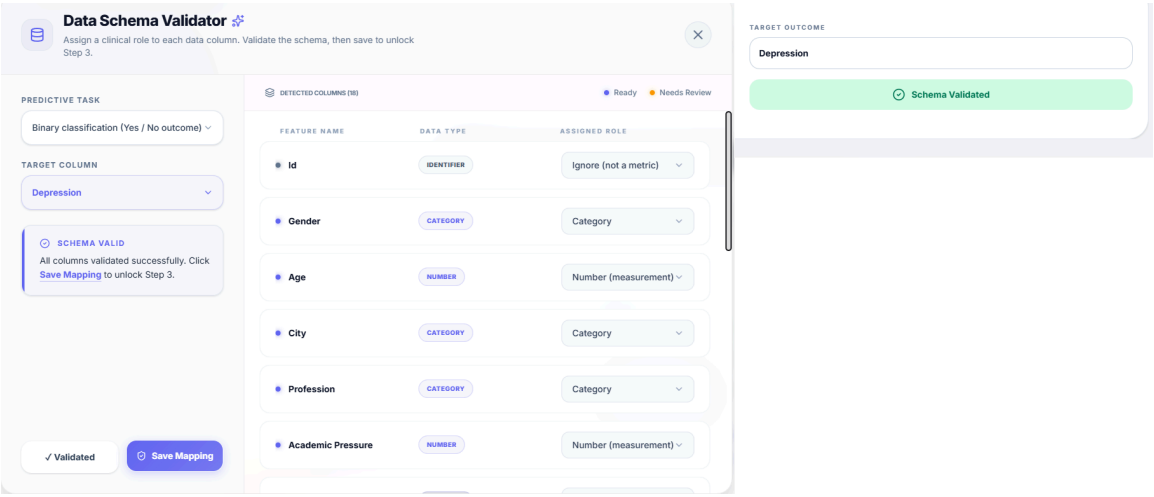
Status: Done



Feedback #7

- Feedback: After clicking 'Save' in the Column Mapper, the success status is not clearly displayed to the user.
- Action: Banner state was fixed; a green success banner with a 2-second fade-out animation was added post-save.

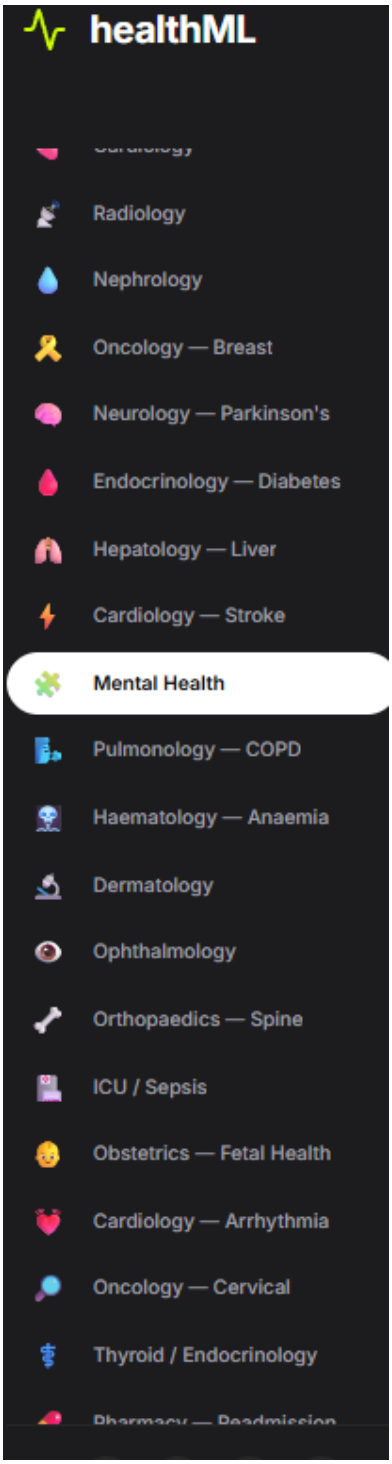
Status: Done



Feedback #8

- Feedback: Clinical sense-check banner texts for the 20 selectable medical domains are incomplete.
- Action: 20/20 domains completed

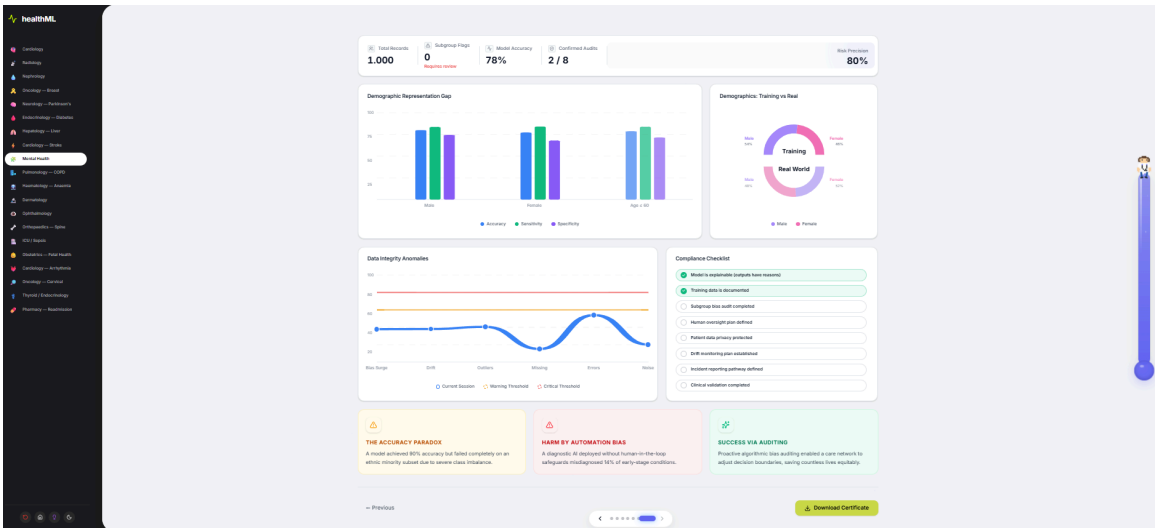
Status: Done



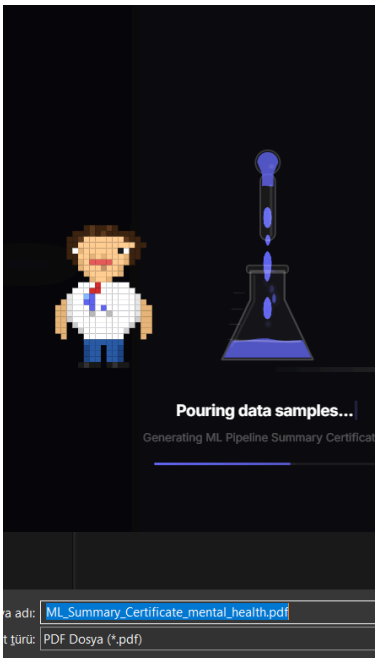
Feedback #9

- Feedback: The bias banner does not trigger at the 11% threshold post-training; boundary test fails.
- Action: Threshold comparison operator was incorrect (used > instead of >=). Fix is currently in progress.

Status: In Progress



Feedback #10

- Feedback: When downloading the Summary Certificate as a PDF, the formatting is misaligned, and charts/tables do not fit on the page.
- Action: CSS margins in the PDF generation function were corrected; tables now scale responsively.

Status: Done